

Software Requirements: Analysis and Specification

Functional and Non-Functional Requirements

- Functional requirements describe the functions that the software is to execute;
 - For example, formatting some text or modulating a signal.
 - They are sometimes known as capabilities or features.
- A functional requirement can also be described as one for which a finite set of test steps can be written to validate its behavior.
- Non-functional requirements are the ones that act to constrain the solution.
 - Non-functional requirements (NFRs) are sometimes known as constraints or quality requirements.
 - They can be further classified according to whether they are performance requirements, maintainability requirements, safety requirements, reliability requirements, security requirements, interoperability requirements or one of many other types of software requirements.

Emergent Properties

- Some requirements represent emergent properties of software—that is, requirements that cannot be addressed by a single component but that depend on **how all the software components interoperate**.
- The throughput requirement for a call center would, for example, depend on how the telephone system, information system, and the operators all interacted under actual operating conditions.
- Emergent properties are crucially dependent on the system architecture.

Requirements Analysis and Specification

- Many projects fail:
 - Because they start implementing the system.
 - Without determining whether they are building what the customer really wants.
- It is important to learn:
 - Requirements analysis and specification techniques carefully.

Requirements Analysis and Specification

- Goals of requirements analysis and specification phase:
 - Fully understand the user requirements.
 - Remove inconsistencies, anomalies, etc. from requirements.
 - Document requirements properly in an SRS document.
- Consists of two distinct activities:
 - Requirements Gathering and Analysis
 - Specification

Who Carries Out Requirements Analysis and Specification?

- The person who undertakes requirements analysis and specification:
 - Generally, the **systems analyst**.
 - Collects data pertaining to the product
 - Analyzes collected data:
 - To understand what exactly needs to be done.
 - Writes the **Software Requirements Specification (SRS)** document.

Requirements Analysis and Specification

- Final output of this phase:
 - Software Requirements Specification (SRS) Document.
- The SRS document is reviewed by the customer.
 - Reviewed SRS document forms the basis of all future development activities.

Requirements Analysis

- Requirements analysis consists of two main activities:
 - Requirements gathering
 - Analysis of the gathered requirements
- Analyst gathers requirements through:
 - Observation of existing systems,
 - Studying existing procedures,
 - Discussion with the customer and end-users,
 - Analysis of what needs to be done, etc.

Requirements Gathering

- Also known as requirements elicitation.
- If the project is to automate some existing procedures
 - e.g., automating existing manual accounting activities,
 - The task of the system analyst is a little easier
 - Analyst can immediately obtain:
 - input and output formats
 - accurate details of the operational procedures

Requirements Gathering Activities

1. Studying the existing documentation
2. Interview
3. Task analysis
4. Scenario analysis
5. Form analysis

Requirements Gathering (CONT.)

- In the absence of a working system,
 - Lot of imagination and creativity are required.
- Interacting with the customer to gather relevant data:
 - Requires a lot of experience.
- Some desirable attributes of a good system analyst:
 - Good interaction skills,
 - Imagination and creativity,
 - Experience.

Analysis of the Gathered Requirements

- Main purpose of requirements analysis:
 - Clearly understand the user requirements,
 - Detect inconsistencies, ambiguities, and incompleteness.
- Incompleteness and inconsistencies:
 - Resolved through further discussions with the end-users and the customers.

Inconsistent Requirement

- Some part of the requirement:
 - contradicts with some other part.
- Example:
 - One customer says turn off heater and open water shower when temperature > 100 DegCel
 - Another customer says turn off heater and turn ON cooler when temperature > 100 DegCel

Incomplete Requirement

- Some requirements have been omitted:
 - Possibly due to oversight.
- Example:
 - Analyst has not recorded what happens when temperature falls below 90 DegCel
 - heater should be turned ON
 - water shower turned OFF.